Problem Set 8 - Solutions

Problem 5.2

A Carnot engine receives 250 kJ s^{-1} of heat from a heat source reservoir at 525°C and rejects heat to a heat-sink reservoir at 50°C. What are the power developed and the heat ejected?

Solution

Use equation 5.7 to compute efficiency, and then use 5.6 to get the work, based on the heat received.

```
In[ ]:= η = 1 - (50 + 273.15)/(525 + 273.15)
Out[ ]=
0.595 126 2
```

```
In[ ]:= QH = 250 * kJ/sec ; (*heat in is positive*)
W = -η * QH (*work out is negative*)
Out[ ]=
-148.7816 kJ/sec
```

The power developed is -148.782 kJ/sec = -148.782 kW. //ANS

To compute the heat ejected, use equation 2.3 (first law) with ΔU=0, that is QC+QH+W=0:

```
In[ ]:= QC = -QH - W
Out[ ]=
-101.2184 kJ/sec
```

Heat rejected is -101.218 kJ/sec = -101.218 kW. //ANS

Problem 5.6

Which is the more effective way to increase the thermal efficiency of a Carnot engine: to increase T_H with T_C constant, or to decrease T_C with T_H constant? For a real engine, which would be the more practical way?

Solution

Efficiency η is a function of two variables, T_C and T_H. That is, $\eta = \eta(T_C, T_H)$:

$$\eta = 1 - \frac{T_C}{T_H}$$

It should be apparent that increasing T_H with T_C held constant increases η, and decreasing T_C with constant T_H causes η to increase.

However, the question is which is more effective? To answer that, we need to look at the rate of change with respect to T_C and T_H. These rates are given by the partial derivatives.

The relationship between the partial derivatives and the total differential is shown below.

$$\mathrm{d}\eta = \left(\frac{\partial \eta}{\partial T_C}\right)_{T_H} \mathrm{dT}_C + \left(\frac{\partial \eta}{\partial T_H}\right)_{T_C} \mathrm{dT}_H$$

The partial derivatives give the slopes of η in the T_C and T_H directions.

$$\left(\frac{\partial \eta}{\partial T_C}\right)_{T_H} = -\frac{1}{T_H}$$

$$\left(\frac{\partial \eta}{\partial T_H}\right)_{T_C} = -T_C\left(-\frac{1}{T_H^2}\right) = \frac{1}{T_H}\,\frac{T_C}{T_H}$$

Compare the magnitude or "size" of the two partial derivatives to see whether T_H or T_C causes a larger change. By premise, $T_C < T_H$, so $\frac{T_C}{T_H} < 1$, and

$$\frac{1}{T_H}\,\frac{T_C}{T_H} < \left|-\frac{1}{T_H}\right|$$

In other words,

$$\left|\left(\frac{\partial \eta}{\partial T_H}\right)_{T_C}\right| < \left|\left(\frac{\partial \eta}{\partial T_C}\right)_{T_H}\right|$$

Since the magnitude of the derivative respect to T_C is larger, decreasing T_C with T_H held constant is "more effective" at increasing thermal efficiency η than increasing T_H with T_C held constant. //ANS

This is a general solution and applies for all $T_C > 0$, all $T_H > 0$, and for $T_H > T_C$.

In practical terms, the situation is the opposite. In a real engine, it is more practical to increase T_H with T_C held constant. //ANS This is because the cold reservoir for a working engine is normally the surrounding air, and T_C is normally the ambient temperature, which is difficult to change by design. T_H can be changed by redesigning the engine to use a different fuel, changing the amount of heat removed from the engine, changing the pressure of the engine, etc.

Problem 5.8

With respect to 1 kg of liquid water:

(a) Initially at 0°C, it is heated to 100°C by contact with a heat reservoir at 100°C. What is the entropy change of the water? What is the entropy change of the heat reservoir? What is ΔS_{total}?

(b) Initially at 0°C, it is first heated to 50°C by contact with a heat reservoir at 50°C, and then heated to 100°C by contact with a heat reservoir at 100°C. What is ΔS_{total}?

(c) Explain how the water might be heated from 0 to 100 C so that $\Delta S_{total} = 0$.

Solution, Part (a)

```
m = 1; (*kg*)
Cp = 4.184; (*kJ/kg*K*)
T1 = 0 + 273.15;(*initial temperature of the water, K*)
T2 = 100 + 273.15;(*reservoir temperature and final temperature of the water, K*)
```

Reservoir:

In[ ]:=
```
Q100 = -m * Cp * (T2 - T1); (*Q50 is the heat leaving the hot reservoir,
which equals the negative of the heat entering the water.  Units are kJ*)
ΔShot = Q100/T2 (*kJ/K*)
```

Out[ ]=
```
-1.121 265
```

The ΔS of the hot reservoir is -1.12123 kJ/K. //ANS

Water:

In[ ]:=
```
ΔSwater = ∫_T1^T2 m * Cp/T dT (*kJ/K*)
```

Out[ ]=
```
1.305 238
```

The ΔS of the water is 1.30524 kJ/K. //ANS

Total:

In[ ]:=
```
ΔStotal = ΔShot + ΔSwater (*kJ/K*)
```

Out[ ]=
```
0.183 973 2
```

The total entropy change is is 0.18397 kJ/K. //ANS

Solution, Part (b)

In[ ]:=
```
m = 1; (*kg*)
Cp = 4.184; (*kJ/kg*degC*)
T1 = 0 + 273.15;(*initial temperature of the water, K*)
T2 = 50 + 273.15;
(*reservoir temperature and final temperature of the water after first step, K*)
T3 = 100 + 273.15;
(*reservoir temperature and final temperature of the water after second step in K*)
```

STEP 1 (0 TO 50°C):

50°C Reservoir:

In[ ]:=
```
Q50 = -m * Cp * (T2 - T1); (*kJ*)
(*heat leaving the reservoir is equal to the heat entering the water*)
ΔS50 = Q50 / T2 (*kJ/K*)
```

Out[ ]=
```
-0.647 377 4
```

Water going from 0 to 50°C:

In[ ]:=
```
ΔSw1 = ∫_T1^T2 (m * Cp) / T dT (*kJ/K*)
```

Out[ ]=
```
0.703 311 6
```

STEP 2 (50 TO 100°C):

100°C Reservoir:

In[ ]:=
```
Q100 = -m * Cp * (T3 - T2); (*kJ*)
(*heat leaving the reservoir is equal to the heat entering the water*)
ΔS100 = Q100 / T3 (*kJ/K*)
```

Out[ ]=
```
-0.560 632 5
```

Water going from 50 to 100°C:

In[ ]:=
```
ΔSw2 = ∫_T2^T3 (m * Cp) / T dT (*kJ/K*)
```

Out[ ]=
```
0.601 926 4
```

TOTAL ENTROPY FOR STEP 1 + STEP 2:

In[ ]:=
```
ΔStotal = ΔS50 + ΔSw1 + ΔS100 + ΔSw2 (*kJ/K*)
```

Out[ ]=
```
0.097 228 23
```

The total entropy change is is 0.097228 kJ/K. //ANS

Solution, Part (c)

The total entropy change for the 1-step process is 0.18397 kJ/K and for the 2-step process, the total entropy change is 0.097228 kJ/K. Note that the 2-step ΔS is slightly more than 1/2 of the 1-step value. Extrapolating, a 3-step process would have a ΔS of slightly more that 1/3 the 1-step value. As the number of steps is increase further, the total entropy change will approach zero, but never quite get there. To get to zero, we would need an infinite number of steps.

Thus, the “reversible” heating of the water (ΔS=0) requires an infinite number of heat reservoirs covering the range of temperatures from 273.15 to 373.15 K, each one exchanging an infinitesimal quantity of heat with the water and raising its temperature by a differential increment. //ANS

Problem 5.10

An ideal gas, C_P = (7/2)R, is heated in a steady-flow heat exchanger from 70 to 190°C by another stream of the same gas, which enters at 320°C. The flow rates of the two streams are the same, and heat losses from the exchanger are negligible.

(a) Calculate the molar entropy changes of the two gas streams for both parallel and countercurrent flow in the exchanger.

(b) What is ΔS_{total} in each case?

(c) Repeat (a) and (b) for countercurrent flow if the heating stream enters at 200°C.

Solution, Part (a)

```
In[ ]:= (*Gas constant and heat capacity*)
R = 8.314; (*J/mol*K*)
Cp = (7 / 2) R;(*J/mol*K*)
```

COUNTERCURRENT FLOW:

The outlet temperature of the hot gas is 200°C:

```
(*Level 1 enthalpy balance gives m·Cp·ΔTc=-m·Cp·ΔTh *)
(*Both m and Cp cancel, so ΔTc=-ΔTh  *)
(*190-70==-(Thout-320) *)
```

```
In[ ]:= Solve[(190 - 70) == -(THout - 320)]
```

```
Out[ ]=
{{THout → 200}}
```

Cold stream:

```
In[ ]:= TCin = 70 + 273.15;(*Cold-side inlet T in K *)
TCout = 190 + 273.15;(*Cold-side outlet T in K *)
```

Use the integrated form of Equation 5.10, with constant C_P and assuming constant pressure:

```
In[ ]:= ΔScold = Cp * Log[TCout / TCin] - R * Log[P / P]
```

```
Out[ ]=
8.726 304
```

The molar entropy change of cold stream gas in co-current (parallel) flow is 8.726 $\frac{J}{mol \cdot K}$. //ANS

Hot stream:

```
In[ ]:= THin = 320 + 273.15;(*Hot-side inlet temperature in K *)
THout = 200 + 273.15;(*Hot-side outlet temperature in K *)
```

In[ ]:=
```
ΔShot = Cp * Log[THout/THin] - R * Log[P/P]
```

Out[ ]=
```
-6.577 388
```

The molar entropy change of hot stream gas in co-current flow is -6.577 $\frac{J}{mol \cdot K}$. //ANS

PARALLEL FLOW:

Since the external temperatures are UNCHANGED, the molar entropy changes of the gas streams in parallel AND counter-current flow are the same. //ANS

Solution, Part (b)

In[ ]:=
```
ΔScold + ΔShot
```

Out[ ]=
```
2.148 916
```

The total molar entropy change of hot and cold gas streams is 2.149 $\frac{J}{mol \cdot K}$ //ANS

Solution, Part (c)

The outlet temperature of the hot gas is 80°C:

In[ ]:=
```
Clear[THout]
```

In[ ]:=
```
(*190-70==-(Thout-200) *)
Solve[(190 - 70) == -(THout - 200)]
```

Out[ ]=
```
{{THout → 80}}
```

Cold stream:

No addition calculations are needed. The cold stream entropy change is the same as part (a) because the cold-stream inlet and outlet temperatures are the same as in part (a).

As in part (a), the molar entropy change of cold stream gas in co-current flow is 8.726 $\frac{J}{mol \cdot K}$. //ANS

Hot stream:

The hot stream temperatures are different, so the entropy change needs to be recalculated:

In[ ]:=
```
(*Hot-side temperatures*)
THout = (80 + 273.15);(*Hot-side inlet temperature in K *)
THin = (200 + 273.15);(*Hot-side outlet temperature in K *)
```

In[ ]:=
```
ΔShot = Cp * Log[THout/THin] - R * Log[P/P]
```

Out[ ]=
```
-8.512 027
```

The molar entropy change of hot stream gas in co-current (parallel) flow is -8.512 $\frac{J}{mol \cdot K}$. //ANS

Total entropy change:

In[ ]:= **ΔScold + ΔShot**

Out[ ]=

0.214 277 4

The total molar entropy change of hot and cold gas streams in co-current (parallel) flow is 0.214 $\frac{J}{mol \cdot K}$.

//ANS